



Assignment-1

Web Engineering

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➤ Submitter To	:	Mr Azhar ALi
➤ Subject	:	Web Engineering
➤ Semester	:	6 th
➤ Section	:	< A >
➤ Department	:	BSCS
➤ Registration No	:	2023-KIU-BS5215

<https://github.com/MuhammadEssa-514/Web-Engineering-Course>

visit above link for repository to check assignment code

<https://muhammadesa514.vercel.app>

visit portfolio

I did this assignment in three ways .

Assignment 1 - Web Engineering

Teacher: Mr Azhar Ali

Submitted By: Muhammad Essa

[Github Repo Link](#)

[Portfolio](#)

❖ Html Basic Structure of boiler plate

❖ `<!DOCTYPE html>`

- ❖ This is a mandatory declaration at the very top of the document that tells the browser the page is written in HTML5. If we don't include this declaration, the browser will assume the page is written in an older version of HTML.

❖ `<html>`

- ❖ The html tag is the root element because it acts as a container for other elements and tells the browser everything inside this tag is html

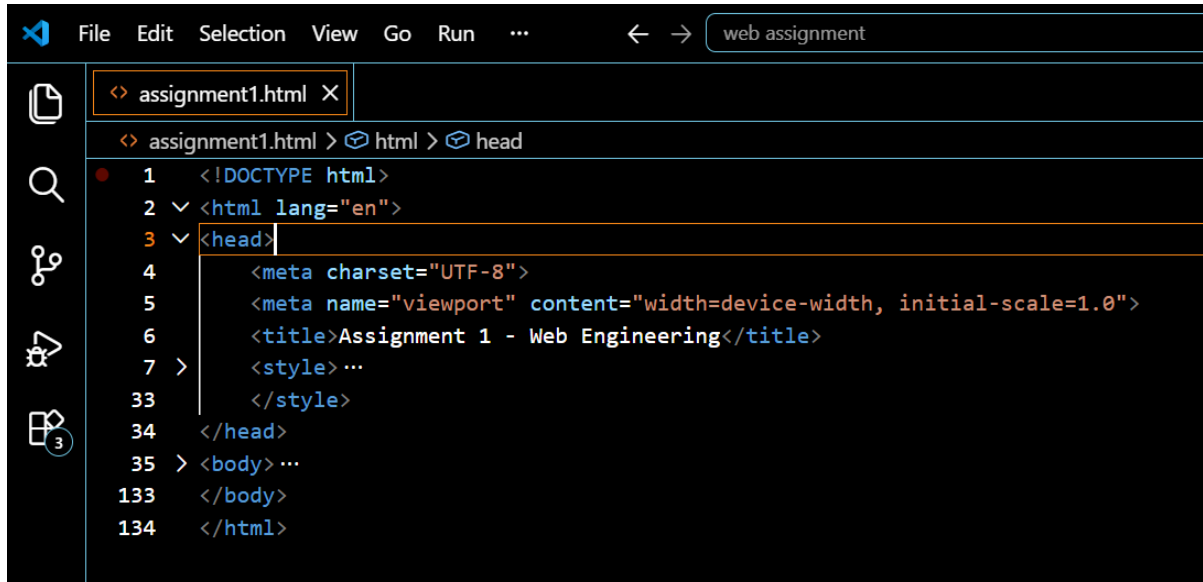
❖ `<head>`

- ❖ A container for metadata (data about the page). This information is not visible to users but is crucial for browsers and search engines.

❖ `<body>`

- ❖ This contains all the visible content, such as text, images, buttons etc.

❖ IDE SS Assignment with simple Css for clear tags visibility



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Assignment 1 - Web Engineering</title>
  <style> ...
</style>
</head>
<body> ...
</body>
</html>
```

❖ 2-Html Web Structure

- **Navbar** contain main links of pages for switching or redirecting to other page or web.
- **Body.** Contain all other content which is visible for users like text,image,videos and buttons etc.
- **footer.** bottom section of web contain social links ,maps ,contact info, copy right notice and privacy policies.

❖ Check basic web structure visit repo for checking practically

HTML Web Structure

Top part of website

HomeAboutContactSignInSignUp

Html Basic Structure of boiler plate

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`<html>`

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`<body>`

This contains all the visible content, such as text, images, buttons etc.

Footer Of Website

❖ Semantic Elements

- ❖ **Semantic Elements**
- ❖ Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

❖ Check SS of Semantic element visit repo for checking practically

Semantic Elements

Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

Header
Main Content
footer
section
paragraph
button

❖ Block Elements

❖ Block Elements:

A block-level element is a type of element that structures a web page by taking up the full width available and always starting on a new line.

Block Elements

A block-level element is a type of element that structures a web page by taking up the full width available and always starting on a new line

<section></section>
<header></header>
<footer></footer>
<main></main>
<nav></nav>
<form action=""></form>
<table></table>
<hr>

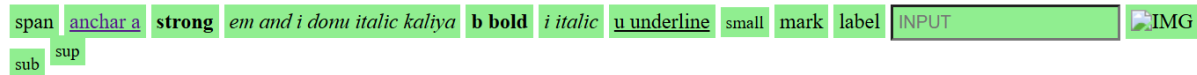
❖ Inline Elements

❖ Inline Elements:

- ❖ An inline element is a type of element that does not start on a new line and only takes necessary space for its content

Inline Elements

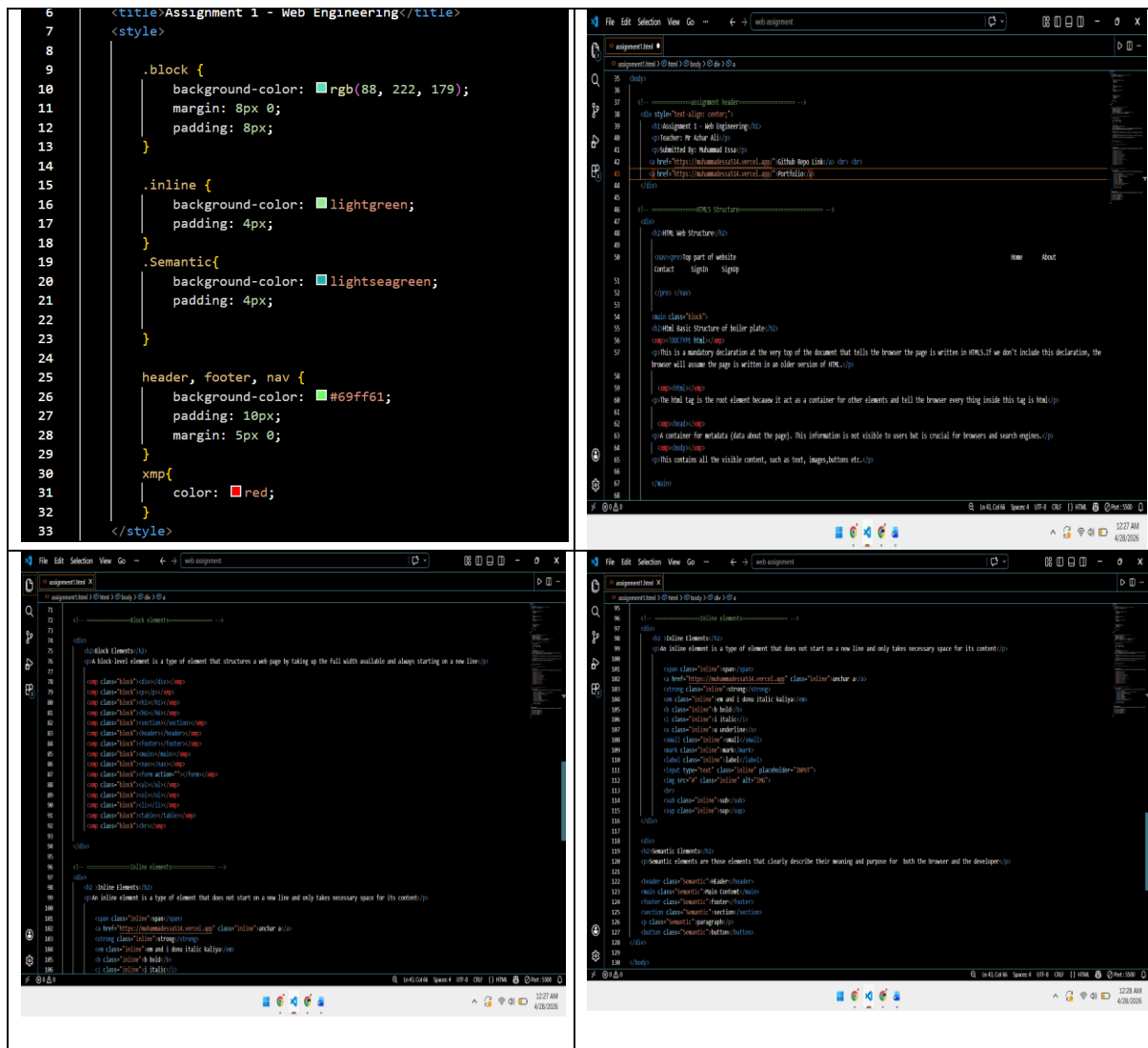
An inline element is a type of element that does not start on a new line and only takes necessary space for its content



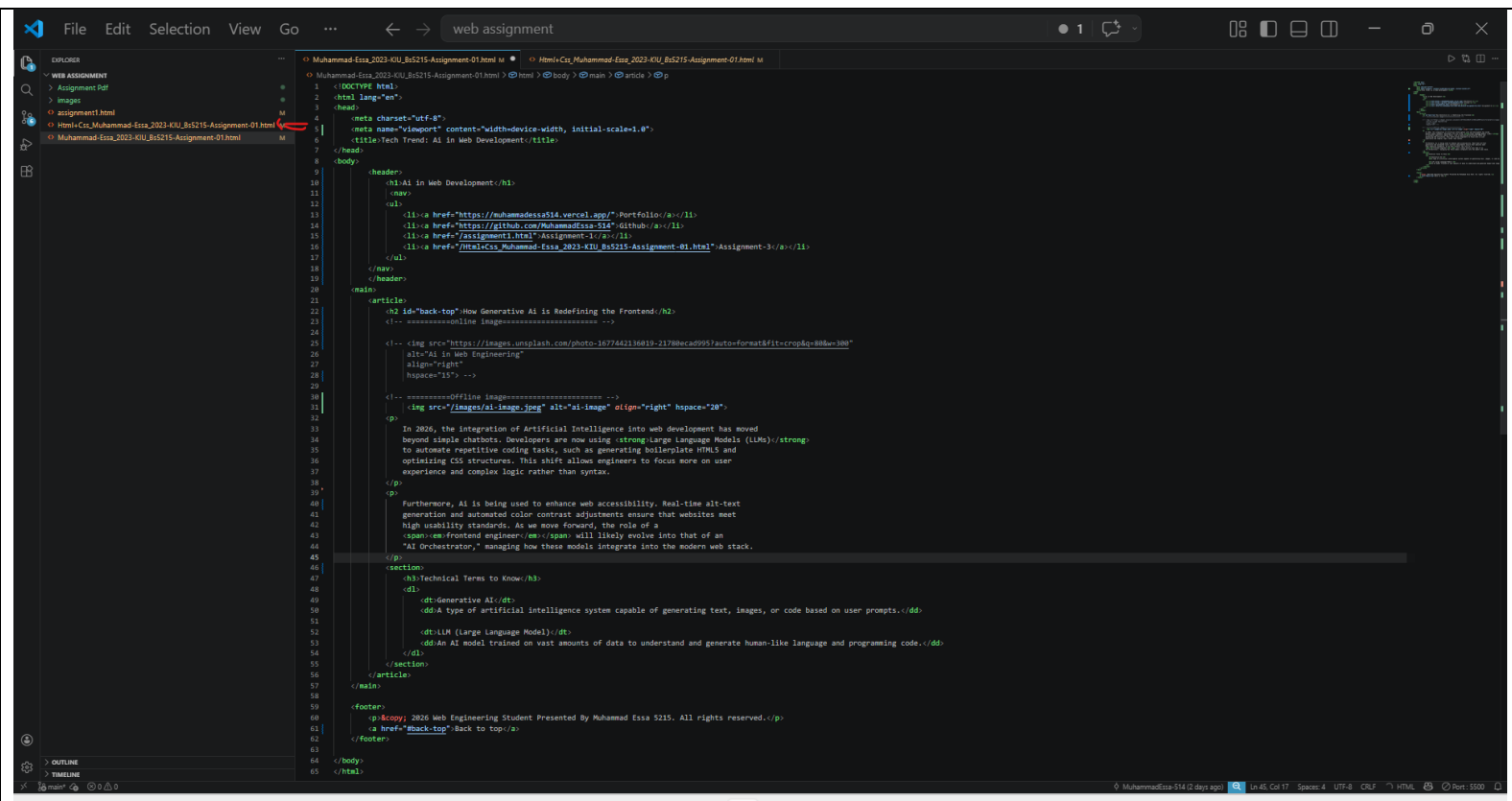
Semantic Elements

Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

Assignment 1 with Simple Css-code

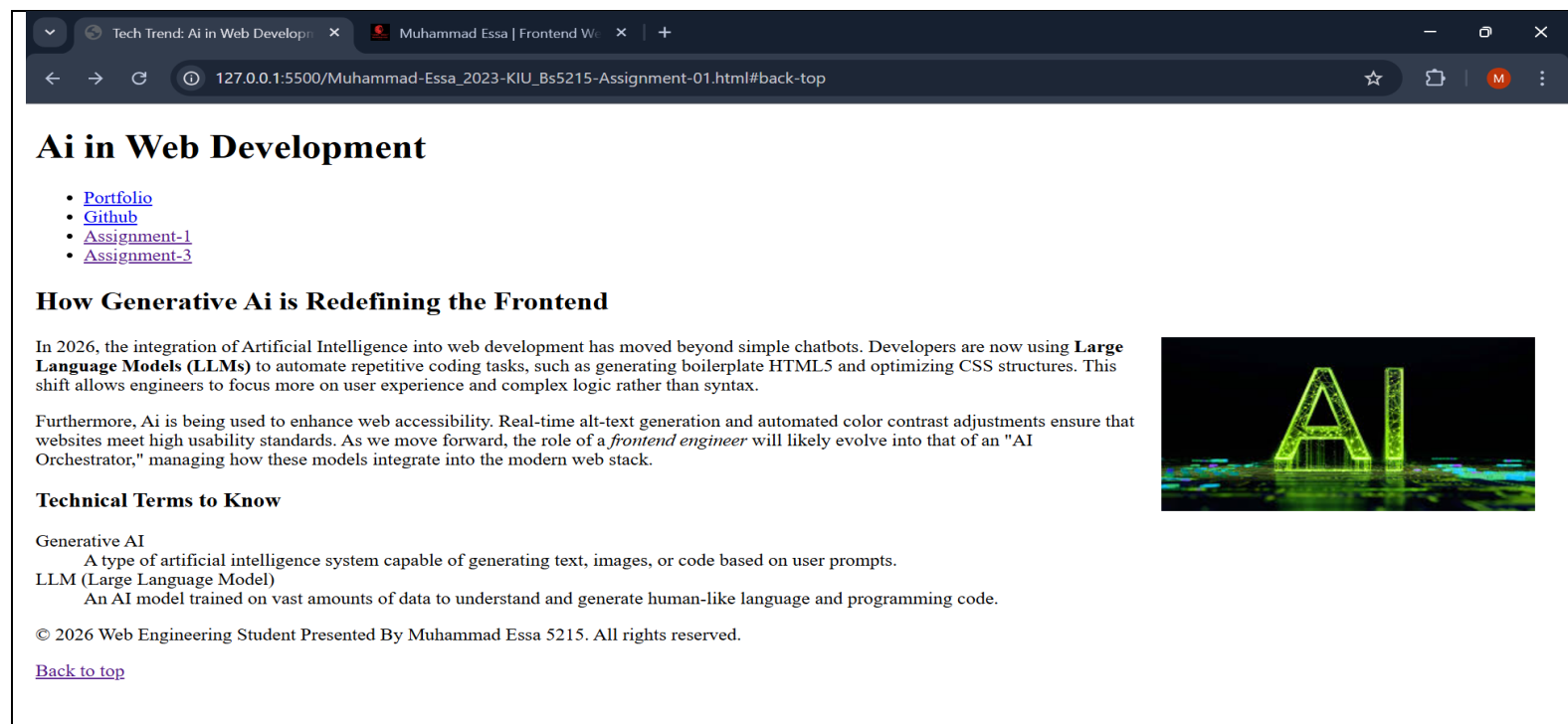


=====Assignment 1 without Css-code=====



```
1 <DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="utf-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Tech Trend: AI in Web Development</title>
7 </head>
8 <body>
9   <header>
10    <h1>AI in Web Development</h1>
11    <div>
12      <ul>
13        <li><a href="https://muhammedessa514.vercel.app"/>Portfolio</a></li>
14        <li><a href="https://github.com/MuhammedEssa514">Github</a></li>
15        <li><a href="/Assignment1.html">Assignment-1</a></li>
16        <li><a href="/html/Css/Muhammed-Essa-2023-KIU_Bs5215-Assignment-01.html">Assignment-3</a></li>
17      </ul>
18    </div>
19    </header>
20   <main>
21     <article>
22       <h2 id="back-top">How Generative AI is Redefining the Frontend</h2>
23       <!-- =====Online Image===== -->
24
25       <!--  -->
29
30       <!-- =====Offline Image===== -->
31       
32
33       <p>
34         In 2026, the integration of Artificial Intelligence into web development has moved
35         beyond simple chatbots. Developers are now using <strong>Large Language Models (LLMs)</strong>
36         to automate repetitive coding tasks, such as generating boilerplate HTML5 and
37         optimizing CSS structures. This shift allows engineers to focus more on user
38         experience and complex logic rather than syntax.
39       </p>
40
41       <p>
42         Furthermore, AI is being used to enhance web accessibility. Real-time alt-text
43         generation and automated color contrast adjustments ensure that websites meet
44         high usability standards. As we move forward, the role of a
45         <span>em>frontend engineer</em></span> will likely evolve into that of an
46         "AI Orchestrator," managing how these models integrate into the modern web stack.
47       </p>
48
49       <section>
50        <h3>Technical Terms to Know</h3>
51        <div>
52          <dt>Generative AI</dt>
53          <dd>A type of artificial intelligence system capable of generating text, images, or code based on user prompts.</dd>
54          <dt>LLM (Large Language Model)</dt>
55          <dd>An AI model trained on vast amounts of data to understand and generate human-like language and programming code.</dd>
56        </div>
57      </section>
58    </article>
59   </main>
60   <footer>
61     <p>© 2026 Web Engineering Student Presented By Muhammad Essa 5215. All rights reserved.</p>
62     <a href="#back-top">Back to top</a>
63   </footer>
64 </body>
65 </html>
```

=Assignment 1 without Css-code Output=



Ai in Web Development

- [Portfolio](#)
- [Github](#)
- [Assignment-1](#)
- [Assignment-3](#)

How Generative Ai is Redefining the Frontend

In 2026, the integration of Artificial Intelligence into web development has moved beyond simple chatbots. Developers are now using **Large Language Models (LLMs)** to automate repetitive coding tasks, such as generating boilerplate HTML5 and optimizing CSS structures. This shift allows engineers to focus more on user experience and complex logic rather than syntax.

Furthermore, Ai is being used to enhance web accessibility. Real-time alt-text generation and automated color contrast adjustments ensure that websites meet high usability standards. As we move forward, the role of a *frontend engineer* will likely evolve into that of an "AI Orchestrator," managing how these models integrate into the modern web stack.

Technical Terms to Know

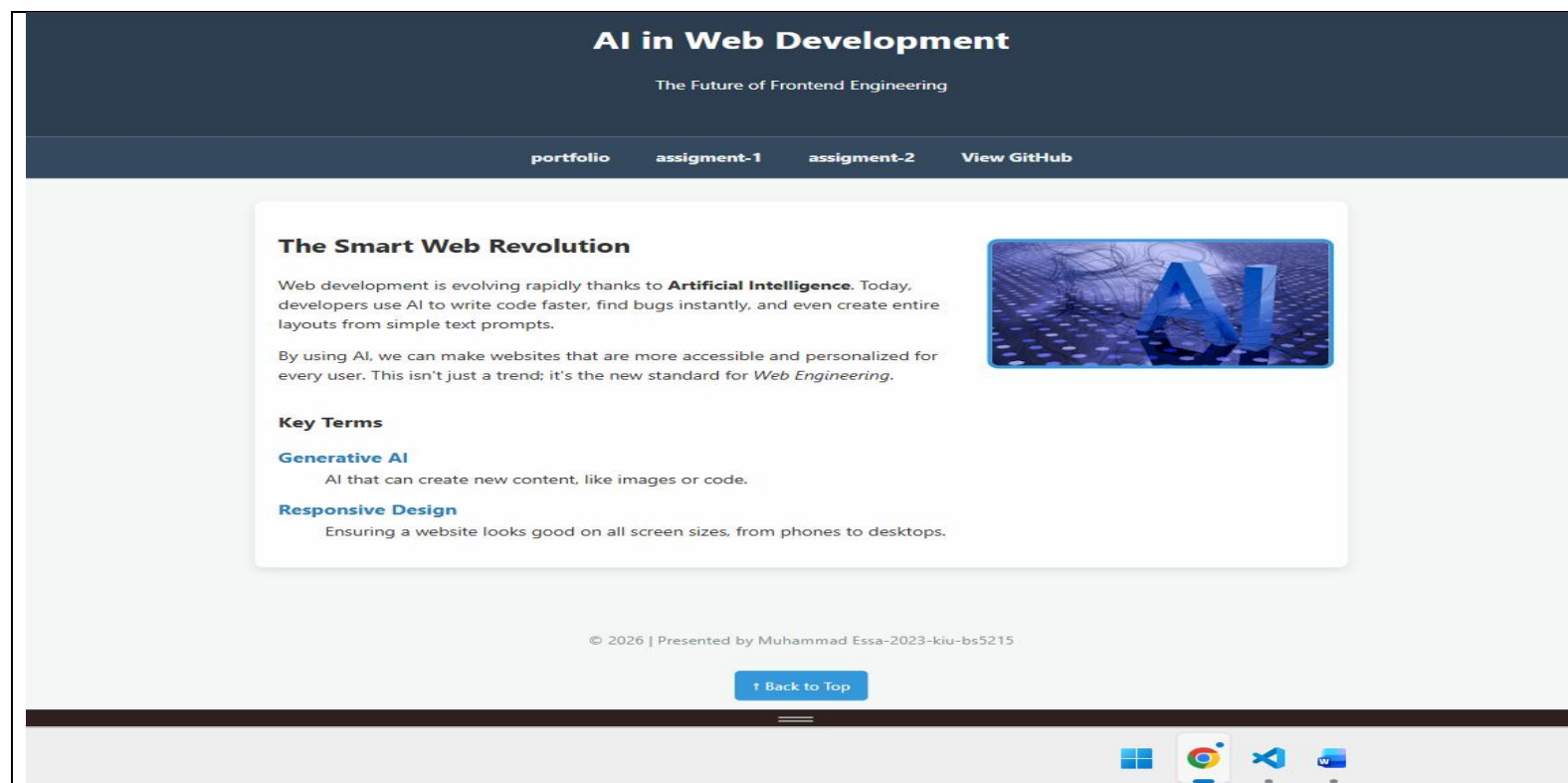
Generative AI
A type of artificial intelligence system capable of generating text, images, or code based on user prompts.

LLM (Large Language Model)
An AI model trained on vast amounts of data to understand and generate human-like language and programming code.

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[Back to top](#)





=Assignment 1 with Proper Css Output=

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="utf-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>AI in Web Development</title>
7 <style>
8 </style>
9 </head>
10 <body id="top"> <!-- Added ID here for the 'Back to top' functionality -->
11 <header>
12 <h1>AI in Web Development</h1>
13 <p>The Future of Frontend Engineering</p>
14 </header>
15 <nav>
16 <ul>
17 <li><a href="https://muhammedessa514.vercel.app/">portfolio</a></li>
18 <li><a href="7assignment1.html">assignment-1</a></li>
19 <li><a href="7Muhammad-Essa_2023-KIU_Bs5215-Assignment-01.html">assignment-2</a></li>
20 <!-- Removed duplicate links for a professional look -->
21 <li><a href="https://github.com" target="_blank">View GitHub</a></li>
22 </ul>
23 </nav>
24 <main>
25 <article id="article">
26 <div class="content-flex">
27 <div class="text-area">
28 <h2>The Smart Web Revolution</h2>
29 <p>
30 Web development is evolving rapidly thanks to <strong>Artificial Intelligence</strong>.
31 Today, developers use AI to write code faster, find bugs instantly, and even
32 create entire layouts from simple text prompts.
33 </p>
34 <p>
35 By using AI, we can make websites that are more accessible and personalized
36 for every user. This isn't just a trend; it's the new standard for
37 <em>Web Engineering</em>.
38 </p>
39 </div>
40 <div class="image-area">
41 
43 </div>
44 </div>
45 <section id="terms">
46 <h3>Key Terms</h3>
47 <dl>
48 <dt>Generative AI</dt>
49 <dd>AI that can create new content, like images or code.</dd>
50 </dl>
51 </section>
52 </main>
53 </body>
54 </html>
```

Thanks Mr Azhar Ali